



University of Sialkot
BSSE BLUE 6th SEMESTER
WEB ENGINEERING
Assignment # 02

Semester Project – Deliverable 1 (Client-Side Architecture)

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TOTAL GRADE POINT	10	
GRADE POINT ALLOCATED		

ASSESSOR'S COMMENTS

Instructor: Museb Khalid

Assignment # 02

Task Requirements & Project Specifications

Project Name: DevPaste – Code Snippets Vue App

1. Single Page Application (SPA) Routing

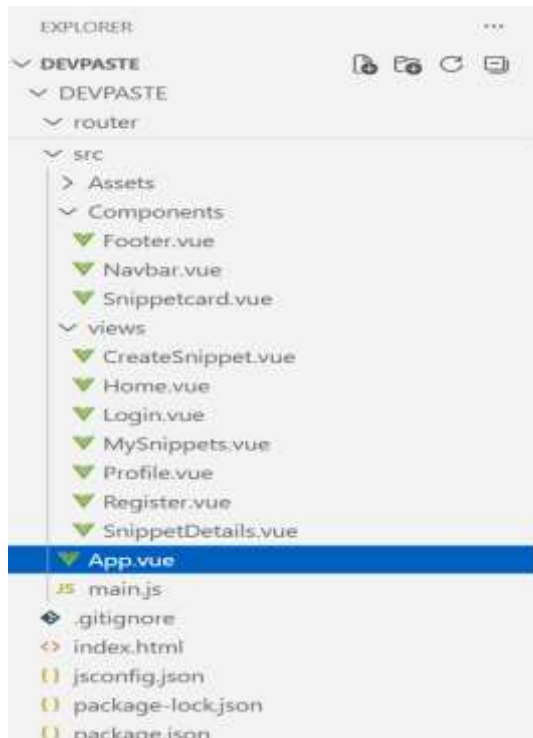
DevPaste uses **Vue Router** to provide smooth navigation between different pages without reloading the browser. Routes are configured for the Home, My Snippets, Create Snippet, Login, Register, and Profile pages. The project has Navbar.vue file that uses the **<router-link>** to add the link of other pages in the navbar.

```
Navbar.vue X
DEV PASTE > src > Components > ▾ Navbar.vue > ...
4  <template>
5    <nav class="navbar navbar-expand-lg navbar-dark bg-dark shadow-sm">
6      <div class="container">
7        </div>
8      <div class="collapse navbar-collapse" id="navbarNav">
9        <ul class="navbar-nav ms-auto">
10         <li class="nav-item">
11           <router-link class="nav-link" to="/">Home</router-link>
12         </li>
13         <li class="nav-item">
14           <router-link class="nav-link" to="/myprofile">My Profile</router-link>
15         </li>
16         <li class="nav-item">
17           <router-link class="nav-link" to="/MySnippets">My Snippets</router-link>
18         </li>
19         <li class="nav-item">
20           <router-link class="nav-link" to="/favorites">Favorites</router-link>
21         </li>
22         <li class="nav-item">
23           <router-link class="nav-link" to="/login">Login</router-link>
24         </li>
25       </ul>
26     </div>
27   </nav>
28 </template>
29
```

App.vue page uses the **<router-view>** to add the dynamic pages or various pages that should occur accordingly.

```
App.vue X
DEV PASTE > src > ▾ App.vue > [ ] template > router-view
1  <script setup>
2    import navbar from './Components/Navbar.vue'
3    import Footer from './Components/Footer.vue'
4  </script>
5
6  <template>
7    <navbar />
8    <router-view />
9    <Footer />
10
11 </template>
12
```

2. Modular Component Architecture



The application is divided into reusable Vue components such as **Navbar**, **Footer**, and **SnippetCard** to improve code organization and maintainability. Each component has a specific responsibility, making the project easier to develop and extend.

3. Inter-Component Communication (Props & Custom Events)

Props are used to pass snippet data from parent components to the **SnippetCard** component. This ensures a one-way data flow while maintaining type safety using `defineProps()`. The **SnippetCard** component uses `defineEmits()` to send events such as **view**, **edit**, and **delete** back to the parent component. This enables the parent to respond to user actions performed on each snippet.

Snippetcard.vue: Defining Props and Emits

```
1 <script setup>
2 defineProps({
3   snippet: {
4     type: Object,
5     required: true
6   }
7 })
8
9 defineEmits(['view', 'edit', 'delete'])
10 </script>
11
12
```

MySnippets.vue: Creating Object

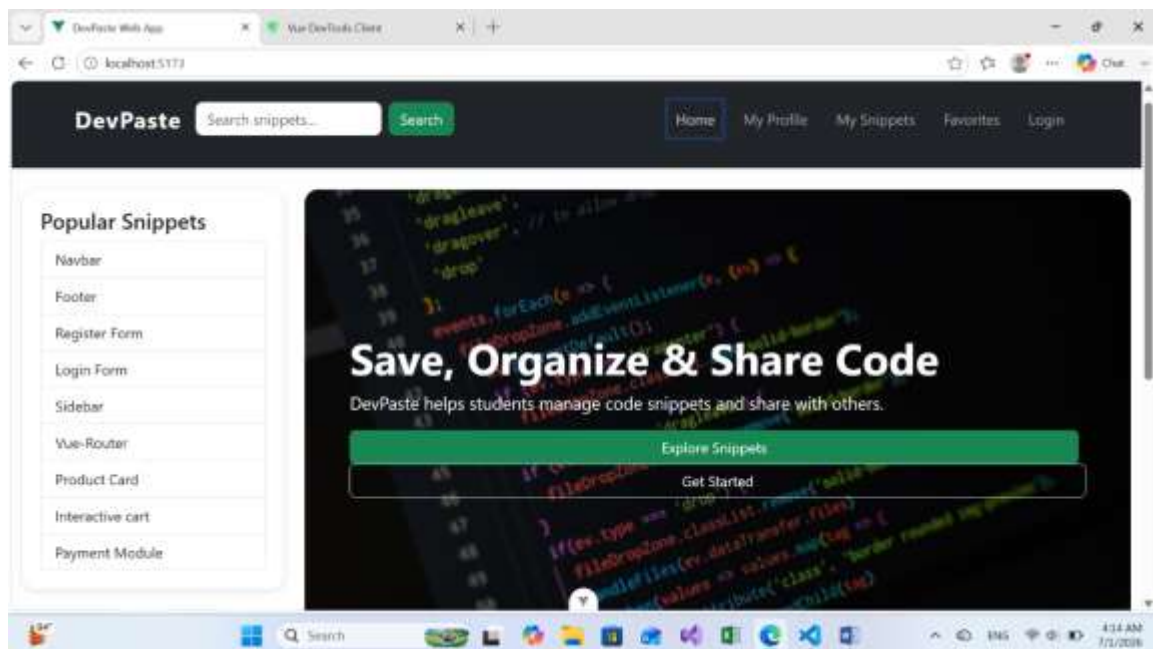
```
>
<SnippetCard
  :snippet="snippet"
  @view="viewSnippet"
  @edit="editSnippet"
  @delete="deleteSnippet"
/>
</div>
```

Snippetcard.vue: Handling the emits

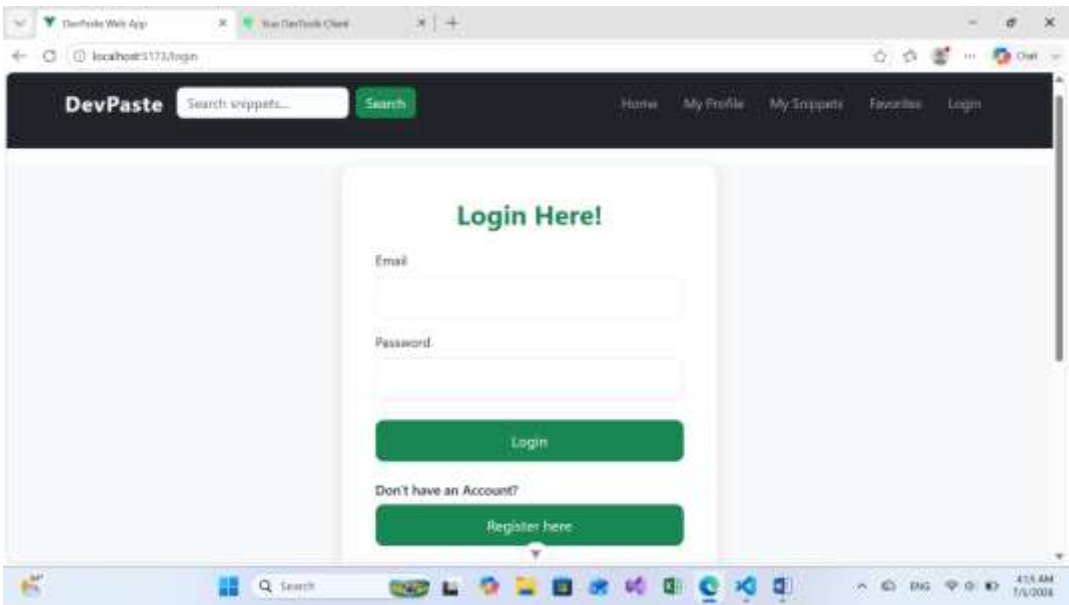
```
44     <div class="d-flex justify-content-between">
45       <button
46         class="btn btn-outline-primary btn-sm"
47         @click="$emit('view', snippet.id)"
48       >
49         View
50       </button>
51
52       <button
53         class="btn btn-outline-warning btn-sm"
54         @click="$emit('edit', snippet.id)"
55       >
56         Edit
57       </button>
58
59       <button
60         class="btn btn-outline-danger btn-sm"
61         @click="$emit('delete', snippet.id)"
62       >
63         Delete
64       </button>
65     </div>
```

Interface Screens from localhost:

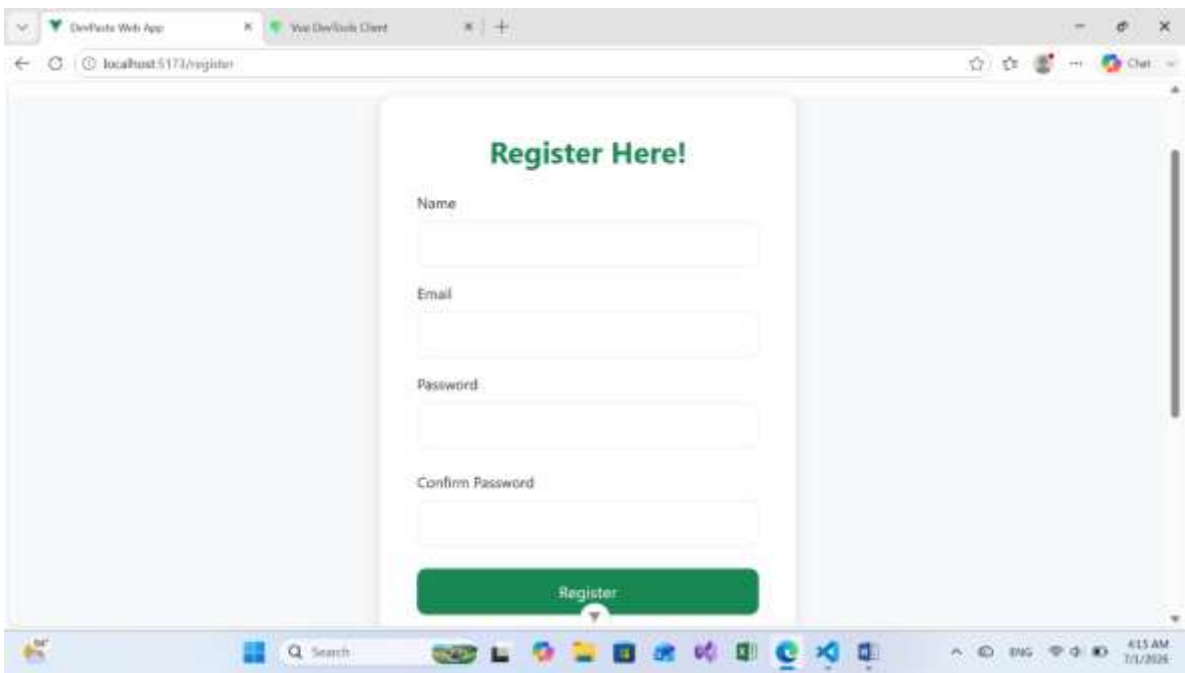
Home page



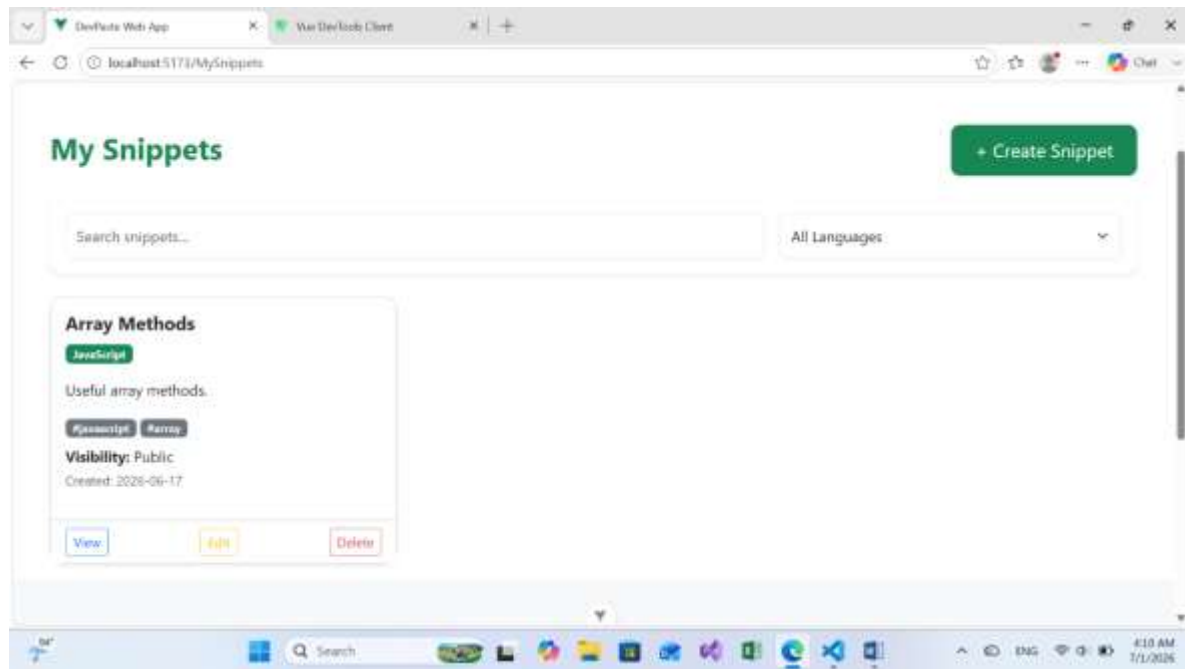
Login Page



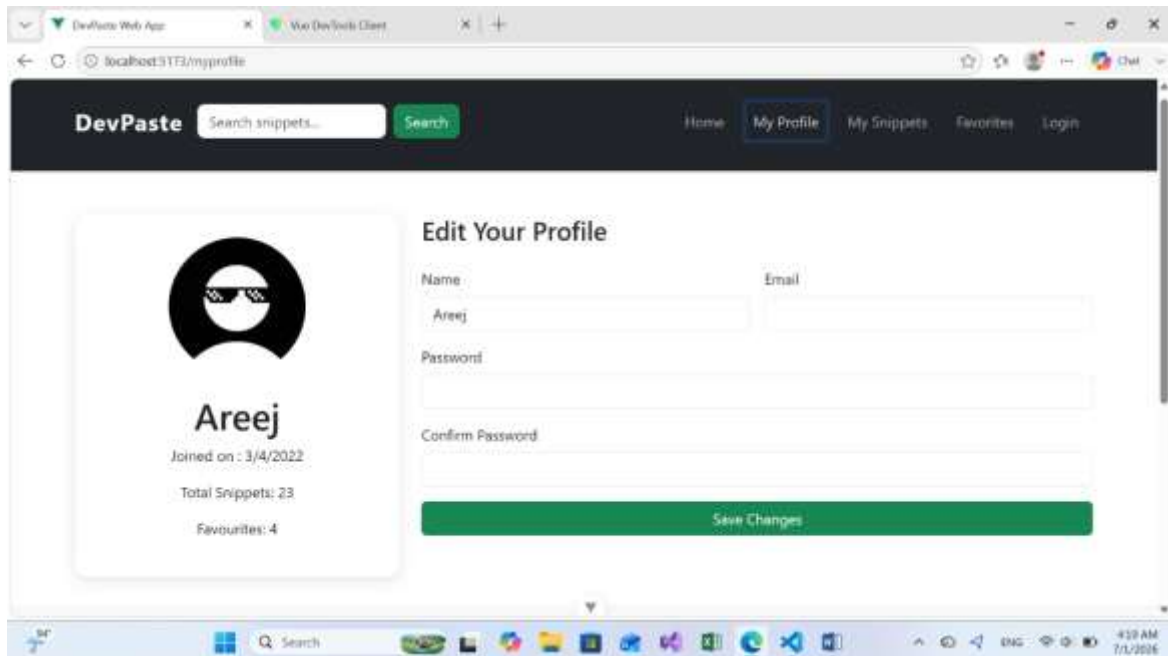
Register Page



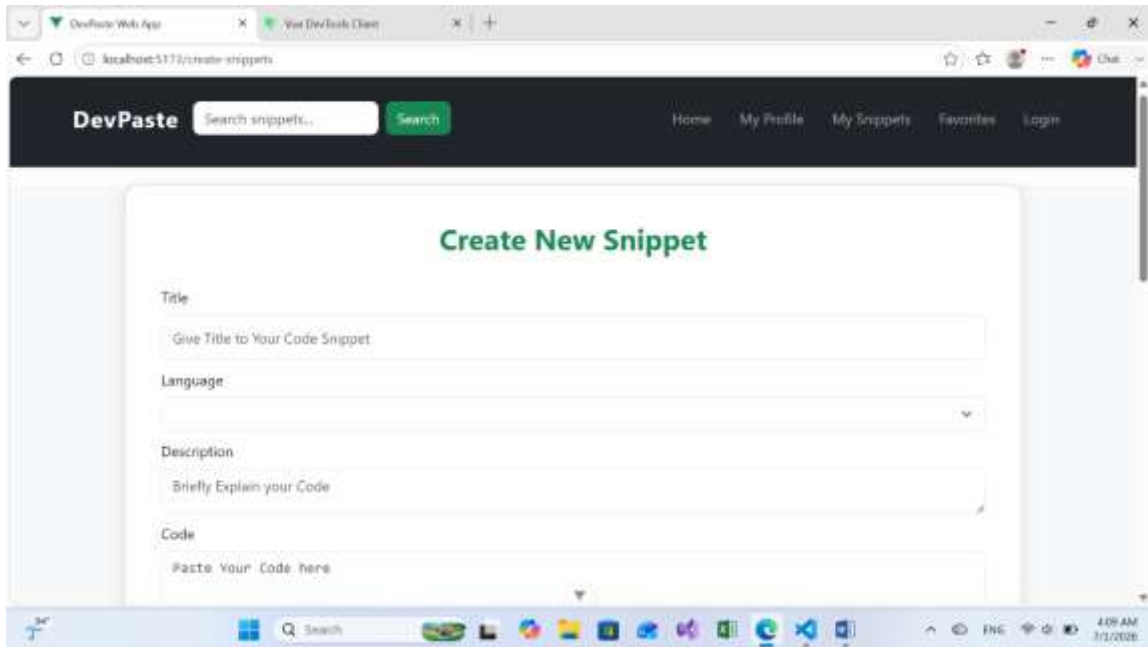
My Snippets Page



Profile page



Create Snippet Page



Project Graph:

